Galois Theory and the Shield Bound: Can Any Polynomial Beat Euler's $p^2 + p + 41$?

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Abstract

Euler's polynomial $p^2 + p + 41$ produces primes at remarkable density: 44.3% of its values at prime arguments $p < 10{,}000$ are prime. We use the Bateman–Horn conjecture and the Chebotarev density theorem to determine what Galois-theoretic conditions a polynomial must satisfy to beat Euler's asymptotic prime production rate.We show that no polynomial of degree ≥ 3 can compete: the "degree tax" in Bateman–Horn requires cubics to block all primes up to ~ 251 (quartics up to ~ 1759), which Chebotarev renders infeasible. Among quadratics, we find that Euler's discriminant $\Delta = -163$ is *not* optimal: the composite discriminant $\Delta = -85507 = -37 \cdot 2311$ achieves a higher Bateman–Horn constant by exploiting ramification to extend the prime-blocking shield. This reveals that the classical Heegner number story is incomplete — composite discriminants with strategically placed ramification can outperform the class-number-one miracle.

Contents

1 Introduction **1**
1.1 Summary of results 2

2 The Bateman–Horn framework **2**

3 Shield bounds by Galois group **3**

4 Euler and the Heegner miracle **3**

5 Beating Euler: composite discriminants **4**
5.1 The ramification mechanism 4
5.2 Crossover analysis 5

6 The cyclic cubic attack **5**

7 Why quadratics win **6**

8 Open questions **6**

1 Introduction

The polynomial $f(x) = x^2 + x + 41$ is one of the most celebrated objects in elementary number theory. Discovered by Euler, it takes prime values for $x = 0, 1, \ldots, 39$, a streak explained by the fact that its discriminant $\Delta = -163$ is the largest Heegner number, corresponding to the imaginary quadratic field $\mathbb{Q}(\sqrt{-163})$ of class number one.

When we evaluate f at prime arguments $p < 10{,}000$, it produces primes 44.3% of the time (545 out of 1229 primes). The Bateman–Horn conjecture predicts that this dominance persists asymptotically, with a constant $C_f/\deg = 3.34$ governing the prime production rate.

The natural question is: *can any polynomial beat Euler?* We bring Galois theory — specifically the Chebotarev density theorem and the fixed-point statistics of Galois groups — to bear on this question.

1.1 Summary of results

(i) For each degree d and transitive Galois group $G \leq S_d$, we compute the minimum *shield bound* B — the largest prime up to which *all* primes must be blocked (have no roots mod p) — required to beat Euler. For $d = 3$ this is $B \approx 263$; for $d = 4$ it is $B \approx 1759$; for $d \geq 5$ it exceeds 10,000. (Section 3.)

(ii) The probability that a random polynomial with Galois group G achieves shield bound B is roughly $\delta(G)^{\pi(B)}$, where $\delta(G)$ is the derangement fraction. For $G = \mathbb{Z}/3\mathbb{Z}$ this is $(2/3)^{56} \approx 10^{-10}$; for all degree-4 and degree-5 groups it is below 10^{-35}. (Section 3.)

(iii) An exhaustive search over negative discriminants Δ down to $-100{,}000$ reveals three quadratics that beat Euler on the Bateman–Horn constant $C_f/2$. The winner is $\Delta = -85507 = -37 \cdot 2311$ with $C_f/2 = 3.48$. (Section 5.)

(iv) The mechanism is *ramification shielding*: the prime 37 divides Δ, making it ramified (contributing a neutral factor of 1 to the Bateman–Horn product), while all primes from 2 to 59 are quadratic non-residues of the cofactor. (Section 5.1.)

(v) The best cyclic cubic found (conductor $q = 4{,}851{,}139$, blocking all primes up to 137) achieves $C_f/3 = 2.88$, still 14% behind Euler. (Section 6.)

2 The Bateman–Horn framework

Let $f \in \mathbb{Z}[x]$ be an irreducible polynomial of degree $d \geq 2$ with positive leading coefficient. The *Bateman–Horn conjecture* [1] predicts

$$\pi_f(N) \;=\; \#\{n \leq N : f(n) \text{ is prime}\} \;\sim\; \frac{C_f}{d} \cdot \mathrm{li}(N),$$

where $\mathrm{li}(N) = \int_2^N dt/\ln t$ is the logarithmic integral and

$$C_f \;=\; \prod_{p \text{ prime}} \frac{1 - \omega_f(p)/p}{1 - 1/p}, \tag{1}$$

with $\omega_f(p) = \#\{x \in \mathbb{Z}/p\mathbb{Z} : f(x) \equiv 0 \pmod p\}$.

The key observation is that C_f/d governs the asymptotic density. To beat Euler, a polynomial of degree d needs $C_f/d > 3.34$.

Definition 1 (Shield)**.** A prime p is *blocked* by f if $\omega_f(p) = 0$ (i.e., f has no roots mod p). The *shield* of f is the set of blocked primes. The *shield product* is $\prod_{p \text{ blocked}} \frac{p}{p-1}$.

When $\omega_f(p) = 0$, the factor at p in (1) is $\frac{p}{p-1} > 1$, boosting C_f. When $\omega_f(p) = k > 0$, the factor is $\frac{p-k}{p-1} < 1$, penalising C_f.

Proposition 2 (Average neutrality)**.** *For any transitive subgroup $G \leq S_d$, the expected value of the Bateman–Horn factor at a prime p (averaged over Frobenius elements by Chebotarev) equals* 1.

Proof. The expected number of fixed points of a uniformly random element of a transitive group is exactly 1 (a standard result in permutation group theory). Therefore

$$\mathbb{E}\left[\frac{1-\omega/p}{1-1/p}\right] = \frac{1}{1-1/p}\left(1-\frac{\mathbb{E}[\omega]}{p}\right) = \frac{1}{1-1/p}\cdot\frac{p-1}{p} = 1. \qquad \square$$

This is fundamental: C_f is not determined by the "average" blocking rate (which is always neutral) but by *which specific small primes* are blocked, since $p/(p-1)$ is largest for small p.

3 Shield bounds by Galois group

For a polynomial f of degree d with Galois group G acting transitively on the d roots, the Chebotarev density theorem determines the distribution of $\omega_f(p)$ over primes:

$$\omega_f(p) = |\operatorname{Fix}(\operatorname{Frob}_p)|,$$

where $\operatorname{Frob}_p \in G$ is the Frobenius element at p (well-defined up to conjugacy for unramified p).

The *derangement fraction* $\delta(G) = |\{g \in G : \operatorname{Fix}(g) = \varnothing\}|/|G|$ gives the asymptotic proportion of blocked primes.

Table 1: Shield bounds required to beat Euler ($C_f/d > 3.34$).

Group G	Degree	$\delta(G)$	Need $C_f >$	Shield B	Feasibility
$\mathbb{Z}/2$	2	50.0%	6.68	37	Euler achieves this
$\mathbb{Z}/3$	3	66.7%	10.02	263	$\sim 10^{10}$ candidates
S_3	3	33.3%	10.02	263	infeasible ($\sim 10^{27}$)
$\mathbb{Z}/4$	4	75.0%	13.36	1759	infeasible ($\sim 10^{35}$)
V_4	4	75.0%	13.36	1759	infeasible ($\sim 10^{35}$)
D_4	4	62.5%	13.36	1759	infeasible ($\sim 10^{56}$)
$\mathbb{Z}/5$	5	80.0%	16.70	11789	infeasible ($\sim 10^{137}$)
F_{20}	5	75.0%	16.70	11789	infeasible ($\sim 10^{177}$)

The shield bound B is the smallest prime such that $\prod_{p \le B} \frac{p}{p-1} > 3.34 \cdot d$. By Mertens' theorem, this product grows as $e^{\gamma} \ln B \approx 1.781 \ln B$.

The "feasibility" column estimates the number of candidates that must be searched: roughly $\delta(G)^{-\pi(B)}$, the reciprocal of the probability that all primes up to B are simultaneously blocked.

Corollary 3 (The degree tax)**.** *For the same shield (blocking all primes up to B), a degree-d polynomial has $C_f/d \approx \frac{1}{d}\prod_{p \le B} \frac{p}{p-1}$. The factor $1/d$ is an inescapable penalty that grows linearly with degree.*

4 Euler and the Heegner miracle

For $f(x) = x^2 + x + 41$, the discriminant is $\Delta = 1 - 4 \cdot 41 = -163$.

A prime p is blocked if and only if $\left(\frac{\Delta}{p}\right) = -1$, i.e., Δ is a quadratic non-residue mod p. By quadratic reciprocity, for odd $p \neq 163$:

$$\left(\frac{-163}{p}\right) = \left(\frac{p}{163}\right),$$

so p is blocked iff p is a QNR mod 163.

The field $\mathbb{Q}(\sqrt{-163})$ has class number $h(-163) = 1$. By the Rabinowitsch criterion, this is equivalent to $x^2 + x + 41$ being prime for all $x = 0, 1, \ldots, 39$. All primes $p < 41$ are QNRs mod 163, giving a shield of 12 consecutive primes.

The shield product is

$$\prod_{p \le 37} \frac{p}{p-1} = \frac{2}{1} \cdot \frac{3}{2} \cdot \frac{5}{4} \cdots \frac{37}{36} \approx 6.72.$$

Beyond $p = 41$, about half of primes remain blocked (by Chebotarev), contributing additional small corrections. The final Bateman–Horn constant is $C_f = 6.68$, giving $C_f/2 = 3.34$.

This connects to deep mathematics:

- Ramanujan's constant: $e^{\pi\sqrt{163}} \approx 640320^3 + 744$ (almost an integer),
- The j-invariant: $j\left(\frac{1+\sqrt{-163}}{2}\right) = -640320^3$,
- Stark–Heegner: there are exactly nine imaginary quadratic fields with class number 1.

5 Beating Euler: composite discriminants

We searched all negative discriminants $\Delta \equiv 0, 1 \pmod 4$ down to $-100{,}000$, computing the Bateman–Horn constant for each.

Table 2: Top quadratics by Bateman–Horn constant $C_f/2$.

Rank	Δ	C_f	$C_f/2$	1st split	Polynomial
1	-85507	6.964	3.482	61	$x^2 + x + 21377$
2	-90787	6.778	3.389	37	$x^2 + x + 22697$
3	-77683	6.705	3.353	47	$x^2 + x + 19421$
4	-163	6.682	3.341	41	$x^2 + x + 41$
5	-86227	6.391	3.196	41	$x^2 + x + 21557$

The winner, $\Delta = -85507$, beats Euler by 4.2% on $C_f/2$.

5.1 The ramification mechanism

The discriminant -85507 factors as $-37 \cdot 2311$, with both 37 and 2311 prime.

- $p = 37$: since $37 \mid \Delta$, the prime is *ramified* in $\mathbb{Q}(\sqrt{-85507})$, giving $\omega_f(37) = 1$ and a Bateman–Horn factor of $\frac{1-1/37}{1-1/37} = 1$ — completely neutral.
- For all primes $p < 61$ with $p \neq 37$: the Kronecker symbol $\left(\frac{-85507}{p}\right) = -1$, so $\omega_f(p) = 0$ (blocked).

This creates a "shield with a neutral hole": the 16 primes $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 43, 47, 53, 59\}$ are blocked, while 37 sits neutrally in the middle. The shield product is ≈ 8.13, compared to Euler's ≈ 7.89.

Remark 4. Euler's polynomial benefits from class number $h(-163) = 1$, which guarantees all primes up to $\sqrt{163/3} \approx 7.4$ are blocked, with the full shield extending to 37 by the specific arithmetic of 163. The discriminant -85507 has class number > 1, but its shield extends further by exploiting ramification to "skip over" the split prime at 37.

5.2 Crossover analysis

Although $\Delta = -85507$ has a higher Bateman–Horn constant, the polynomial $x^2 + x + 21377$ produces larger values (the constant term is 520 times bigger), making individual values less likely to be prime at moderate arguments.

Table 3: Prime counts at prime arguments: Euler vs. $\Delta = -85507$.

N	Euler primes	-85507 primes	Euler density	-85507 density
1 000	109	110	0.649	0.655
2 000	168	176	0.555	0.581
5 000	329	320	0.492	0.478
10 000	545	531	0.443	0.432
20 000	917	907	0.405	0.401
50 000	1885	1901	0.367	0.370

The two polynomials trade the lead repeatedly. By $N = 50{,}000$, the Bateman–Horn prediction begins to dominate: $\Delta = -85507$ pulls ahead with 1901 vs. 1885 primes. The asymptotic advantage of $\sim$4% should become decisive for large N.

6 The cyclic cubic attack

The most promising non-quadratic avenue is cyclic cubics ($G = \mathbb{Z}/3$), which block 2/3 of primes — the highest derangement fraction among degree-3 groups.

For a cyclic cubic with prime conductor q (where $3 \mid q - 1$), the splitting of a prime p is controlled by cubic reciprocity:

$$\omega_f(p) = \begin{cases} 3 & \text{if } p^{(q-1)/3} \equiv 1 \pmod{q} \text{ (cubic residue)}, \\ 0 & \text{otherwise.} \end{cases}$$

We searched all primes $q \equiv 1 \pmod 3$ up to 10^7, looking for conductors where all small primes are cubic non-residues.

Table 4: Best cyclic cubic conductors.

Conductor q	1st cubic residue	# blocked	Shield	$C_f/3$	Density
4,851,139	137	32	8.850	2.878	0.339
7,778,509	131	31	8.782	2.862	0.359
6,843,919	127	30	8.713	2.934	0.356
2,206,591	113	29	8.636	2.964	0.371

The best cubic blocks all primes up to 137 — an impressive shield — but $C_f/3 \approx 2.9$ remains 14% below Euler's 3.34.

For each conductor, we constructed explicit cubic polynomials via Gauss periods. For conductor q, let g be a primitive root mod q and partition $(\mathbb{Z}/q\mathbb{Z})^\times$ into three cosets of the subgroup of cubes. The Gauss period $\eta_0 = \sum_{k \in C_0} \cos(2\pi k/q)$ is a root of a monic cubic with integer coefficients.

For example, conductor $q = 2{,}206{,}591$ yields $x^3 + x^2 - 735{,}530x + 227{,}687{,}501$, a cyclic cubic blocking all 25 primes up to 97.

7 Why quadratics win

We can now state the structural result:

Theorem 5 (Informal)**.** *Under the Bateman–Horn conjecture, no polynomial of degree $d \geq 3$ can match the asymptotic prime production rate of the optimal quadratic. Among quadratics, the optimal discriminant (by Bateman–Horn constant) is not a Heegner number but a composite discriminant exploiting ramification.*

The argument rests on three pillars:

1. The degree tax. The Bateman–Horn prediction scales as C_f/d. For the same shield product S, a degree-d polynomial achieves $C_f/d \approx S/d$. To match Euler's $S/2 \approx 3.34$, degree d needs $S > 3.34 \cdot d$, which by Mertens' theorem requires blocking all primes up to $\exp(3.34d/e^\gamma) \approx \exp(1.88d)$. This grows exponentially with d.

2. Chebotarev makes large shields improbable. By Proposition 2, only a fraction $\delta(G)$ of primes are blocked. The probability that *all* primes up to B are blocked is roughly $\delta(G)^{\pi(B)}$, which decreases super-exponentially with B. Even the best group ($\mathbb{Z}/5$ with $\delta = 4/5$) cannot achieve the shield needed at degree 5.

3. Ramification gives quadratics an extra trick. Composite discriminants allow specific primes to be ramified (neutral) rather than split (penalised), effectively "skipping over" bad primes in the shield. This trick is unique to the interplay of the discriminant's factorisation with quadratic reciprocity, and has no direct analogue for cubic or higher reciprocity that would overcome the degree tax.

8 Open questions

1. **Optimal composite discriminants.** Is there a theoretical framework, analogous to Heegner's theorem for class number one, that predicts which composite discriminants maximise C_f? The interplay of ramification and quadratic reciprocity at composite moduli is rich and, to our knowledge, not well characterised for this optimisation problem.

2. **Deeper cubic search.** Our search covered conductors up to 10^7, finding shields up to $B = 137$. A search to 10^{10} might find shields up to $\sim$200, but $C_f/3$ would need to exceed 3.48 (the new quadratic champion), requiring a shield product above 10.4 — blocking all primes up to $\sim$290.

3. **Function field analogue.** Over $\mathbb{F}_q[t]$, the Bunyakovsky conjecture is a theorem (via Weil's proof of the Riemann hypothesis for curves). Does the proof structure illuminate which polynomials over $\mathbb{Z}$ should produce the most primes?

4. **Non-monic quadratics.** We tested $f(x) = ax^2 + 1$ for various primorial values of a. Even $a = 30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ achieves only $C_f/2 = 2.26$. The "free" blocking of primes dividing a cannot compensate for the poor residual discriminant $-4a$. A more systematic search over general $ax^2 + bx + c$ might be worthwhile.

References

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